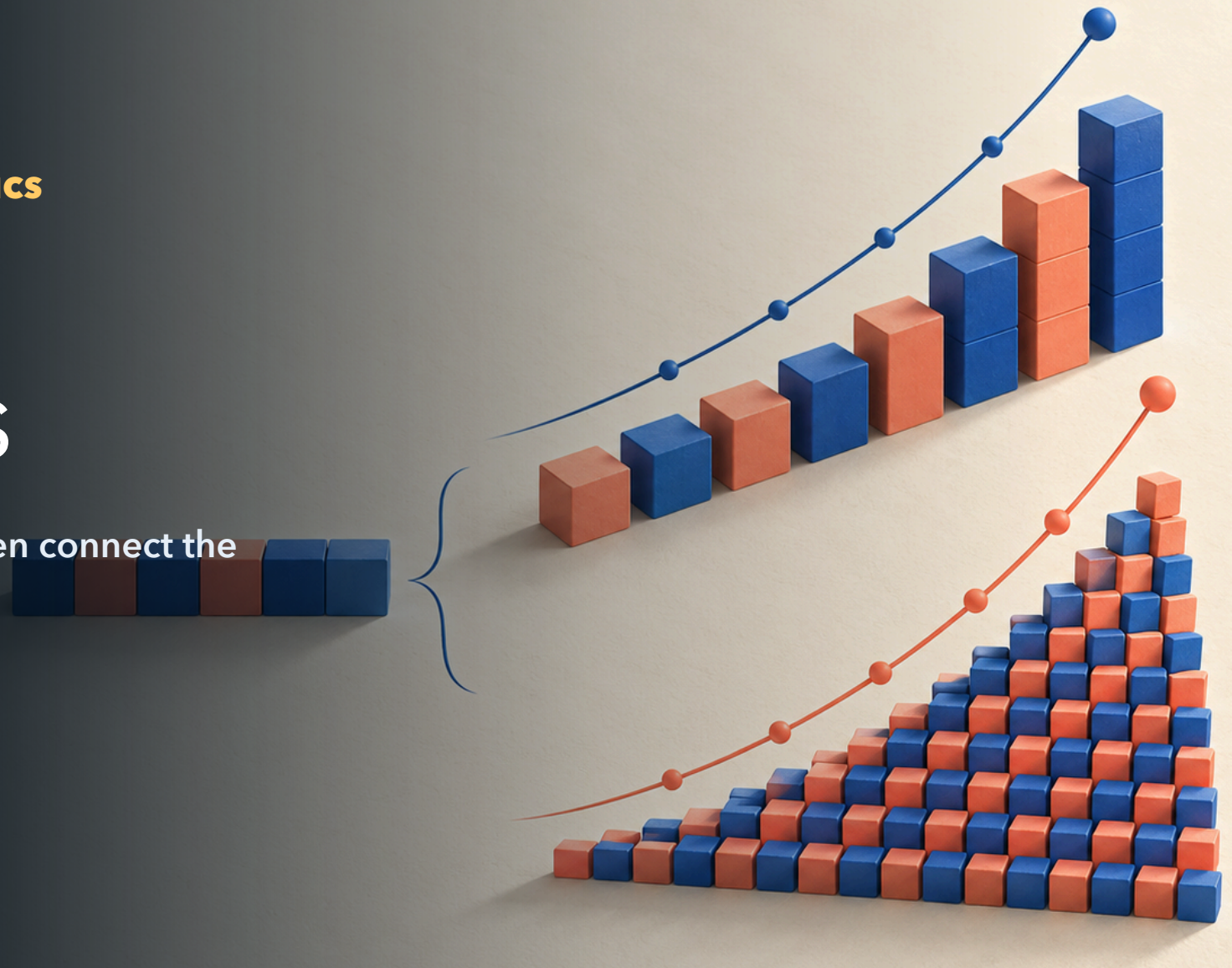
LECTURE 20 · DISCRETE MATHEMATICS

Solving recurrences

Tree method and substitution. Then connect the result to runtime.



READ THE RULE

Identify the three parts of a recurrence

Branching factor

2 copies

Nonrecursive part

Add 1

$$a_n = 2\,a_{n-1} + 1$$

Recursive call

Subproblem size: $n - 1$

For $n \geq 2$. Stop expanding at $a_1 = 2$.

Expand a_5 one step at a time

Start with the term we want to find.

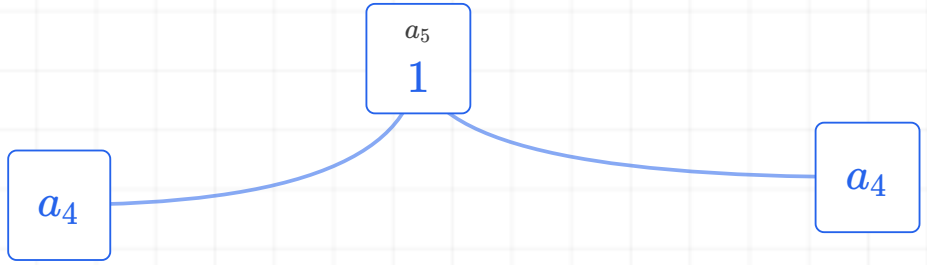
$$a_5 = 1 + a_4 + a_4$$

Record the nonrecursive part

Each expanded a_5 contributes 1.

Branching factor 2

$1 \times 2 = 2$ copies of a_4



2 copies of a_4 now remain.

The 2 endpoint boxes are still terms to resolve.

EXERCISE 1 · CHECKPOINT · 45 SECONDS

Expand a_5 one step at a time

Expand both a_4 boxes once.

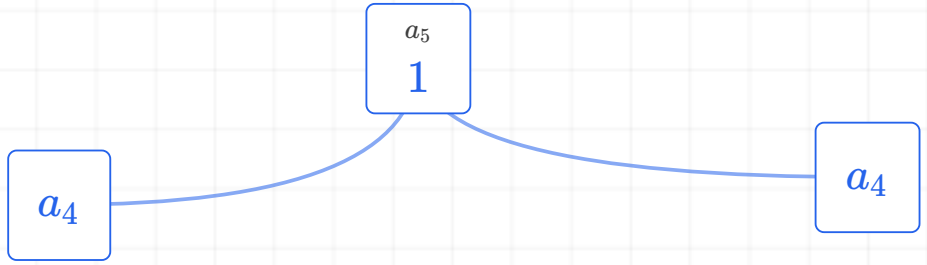
$$a_n = 2a_{n-1} + 1$$

01 **Value left in each old box?**

1

02 **New index? Total new boxes?**

Index 3; 4 new boxes.



Expand a_5 one step at a time

Expand each of the 2 remaining copies of a_4.

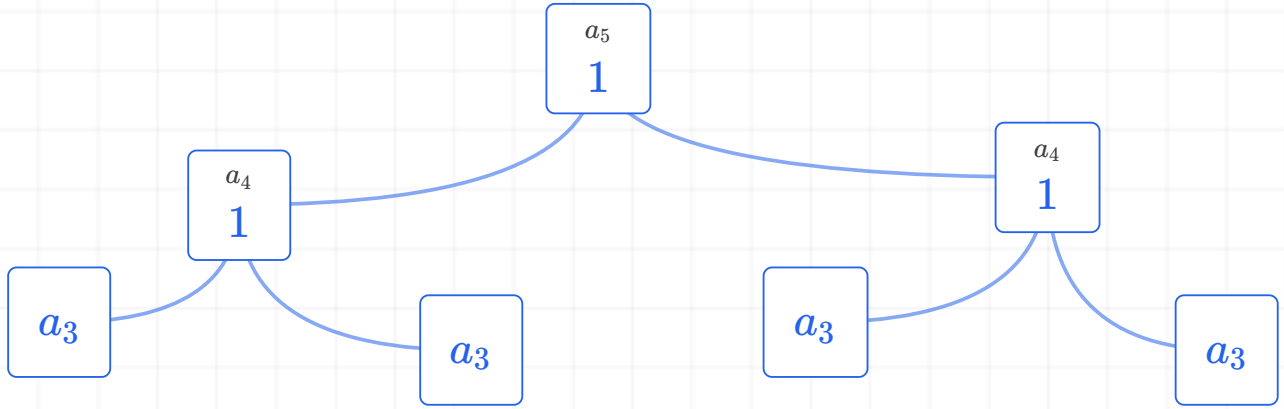
$$a_4 = 1 + a_3 + a_3$$

Record the nonrecursive part

Each expanded a_4 contributes 1.

Branching factor 2

$2 \times 2 = 4$ copies of a_3



4 copies of a_3 now remain.

The 4 endpoint boxes are still terms to resolve.

Expand a_5 one step at a time

Expand each of the 4 remaining copies of a_3.

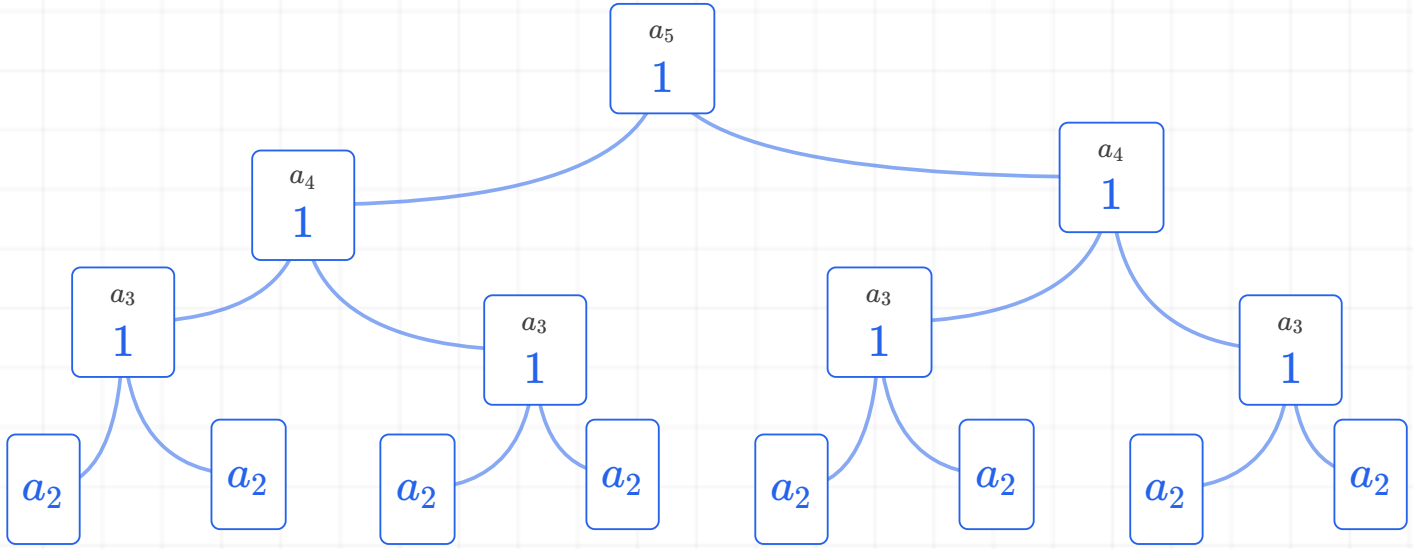
$$a_3 = 1 + a_2 + a_2$$

Record the nonrecursive part

Each expanded a_3 contributes 1.

Branching factor 2

$4 \times 2 = 8$ copies of a_2



8 copies of a_2 now remain.

The 8 endpoint boxes are still terms to resolve.

Expand a_5 one step at a time

Expand each of the 8 remaining copies of a_2.

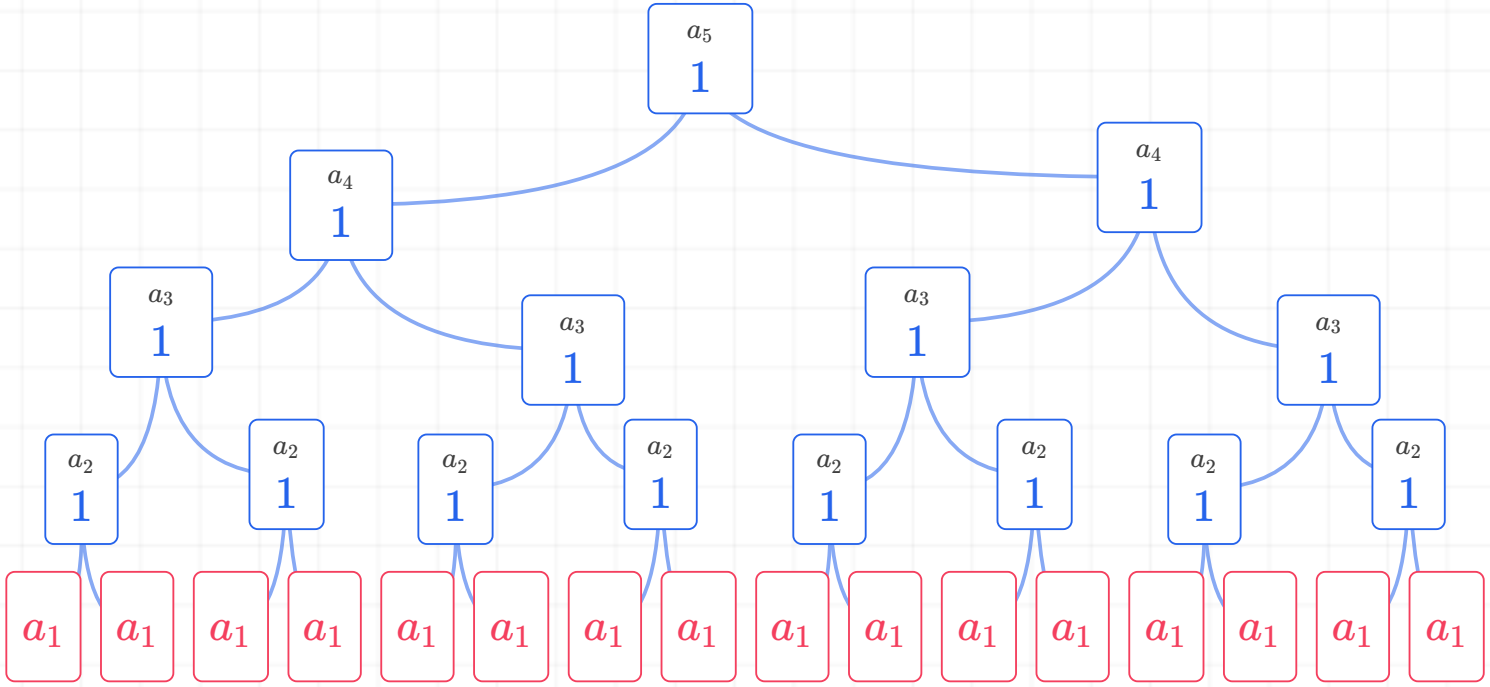
$$a_2 = 1 + a_1 + a_1$$

Record the nonrecursive part

Each expanded a_2 contributes 1.

Branching factor 2

$8 \times 2 = 16$ copies of a_1



16 copies of a_1 now remain.

Every unresolved term has reached index 1.

Expand a_5 one step at a time

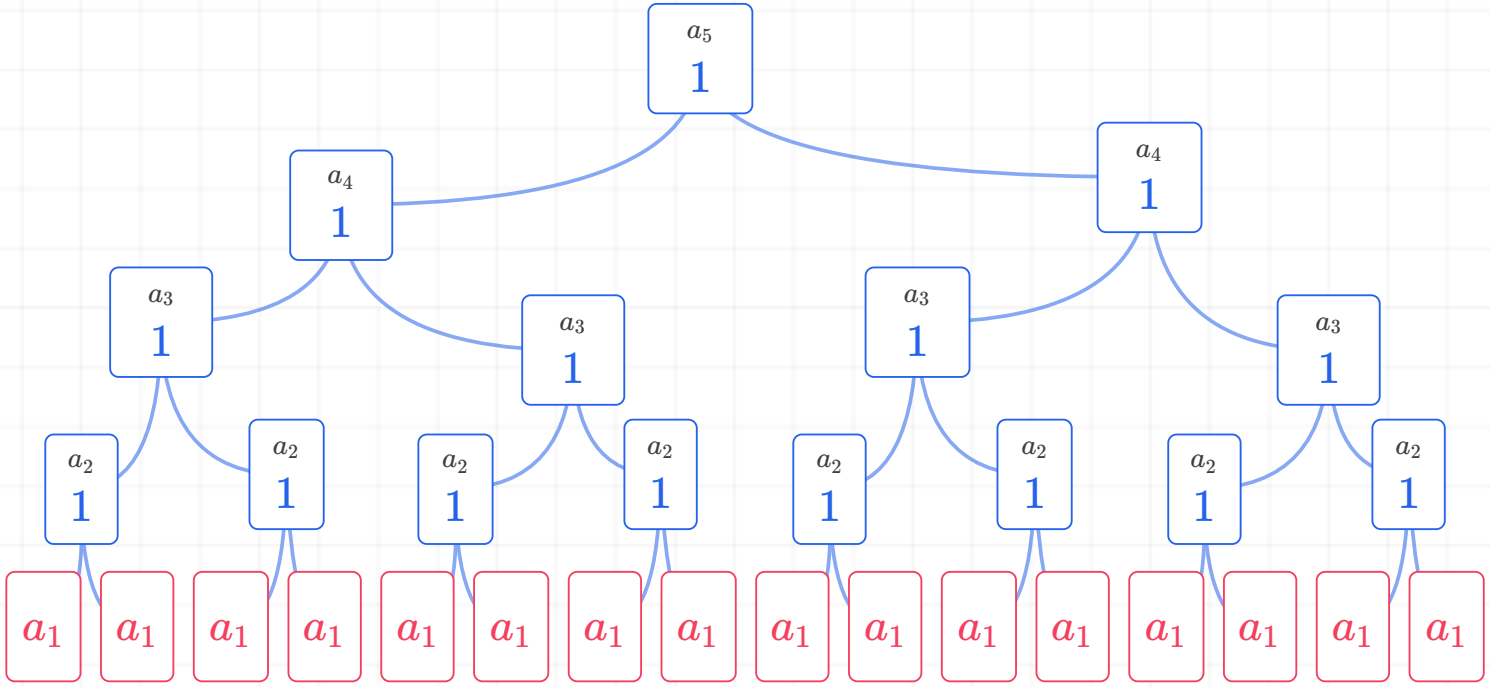
At each endpoint, use $a_1 = 2$.

01 **Value in each endpoint?**

2

02 **New links from each endpoint?**

0 **– stop at the base.**



Expand a_5 one step at a time

The sixteen remaining terms are a_1 . Their value is given.

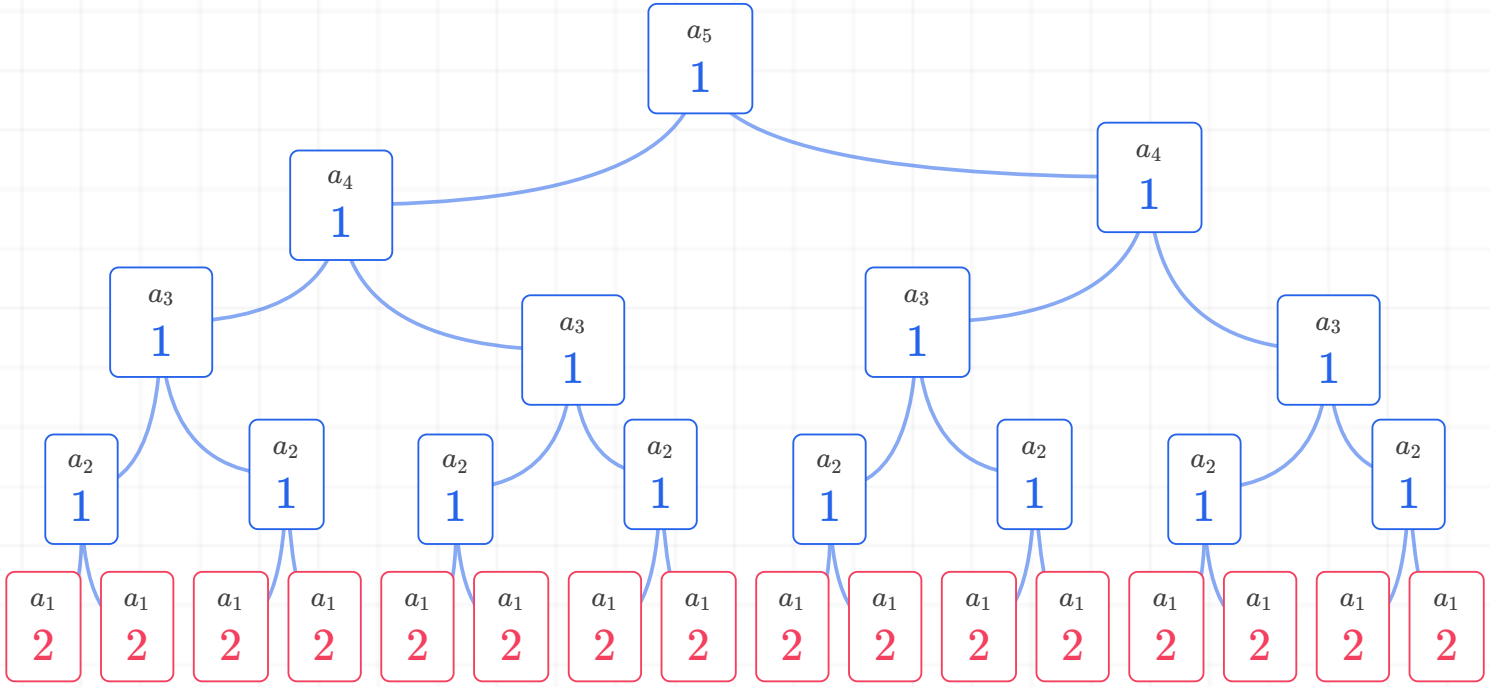
$$a_1 = 2$$

Replace each a_1 with 2

These boxes are complete. They need no more links.

Now every contribution is known

Add every node value exactly once.



Now a_5 is the sum of all 31 numerical node values.

ORGANIZE THE SAME PICTURE

Move the same nodes into layers

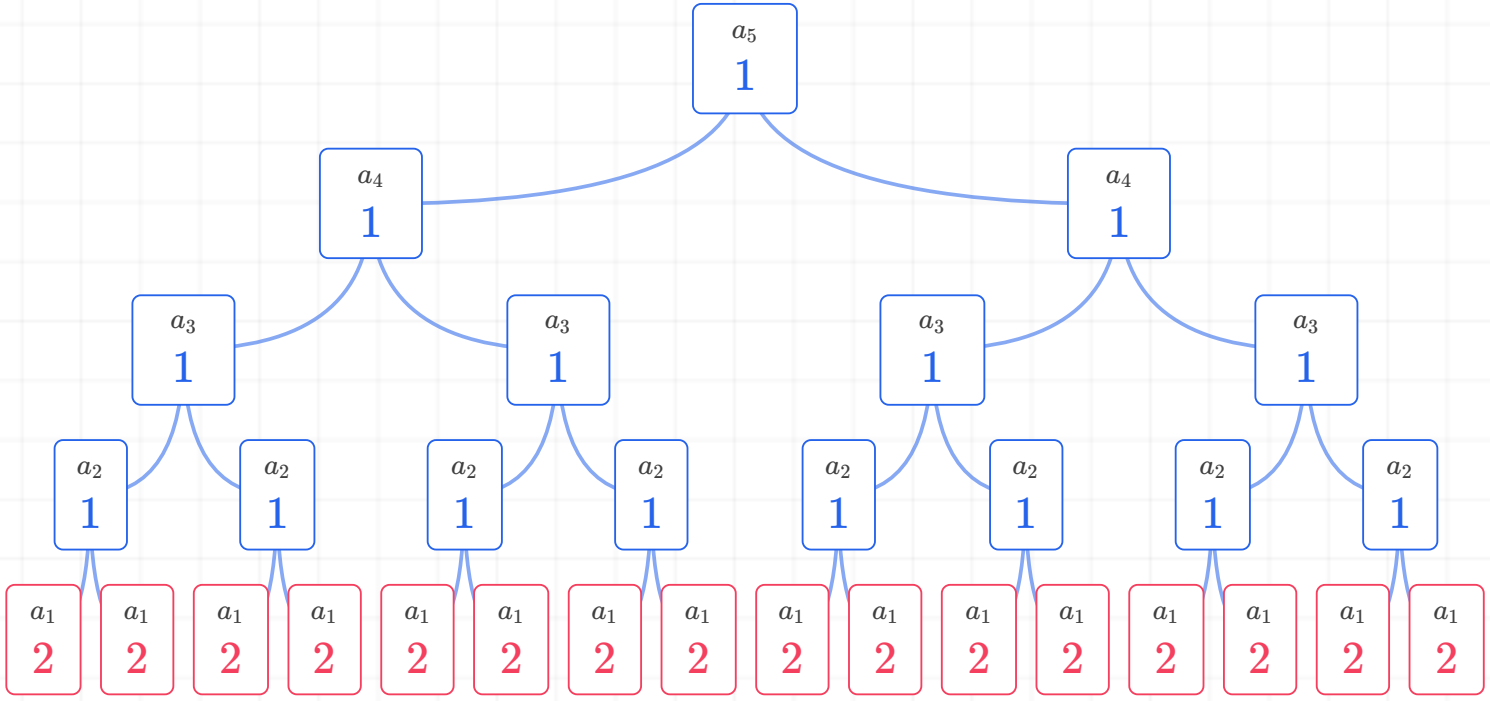
Keep all 31 nodes and all 30 links.
Group them by expansion step.

Top row

The original term.

Next rows

Terms one expansion away, then two expansions away.



One row for each expansion step.

Changing positions does not change the sum.

EXERCISE 3 · CHECKPOINT · 1 MINUTE

Move the same nodes into layers

Use the finished picture.

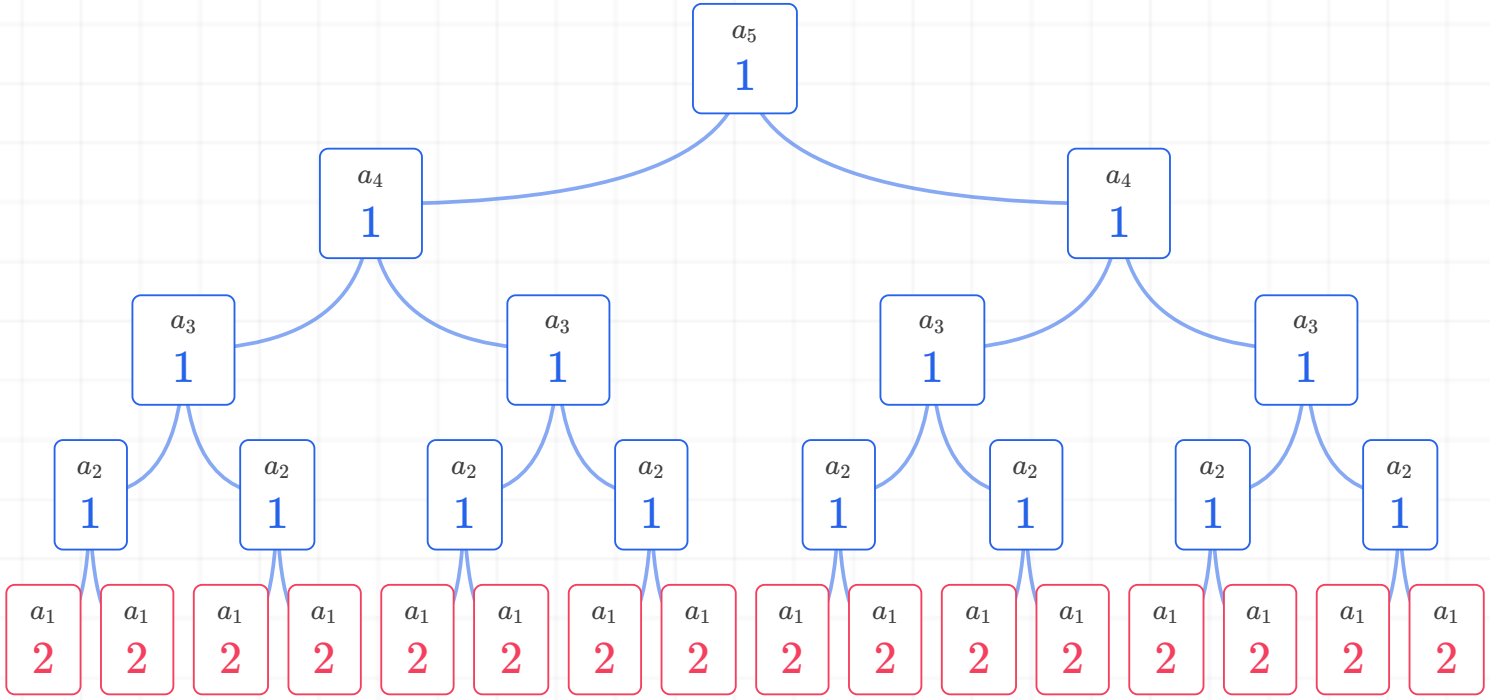
01 **Sum of the 1s? Sum of the 2s?**

15 **and** 32**.**

02 **What is a_5?**

$a_5 = 15 + 32 = 47$

.



Add one layer at a time

This branching picture is a tree. The first node is the root; the endpoints are leaves.

Internal layers

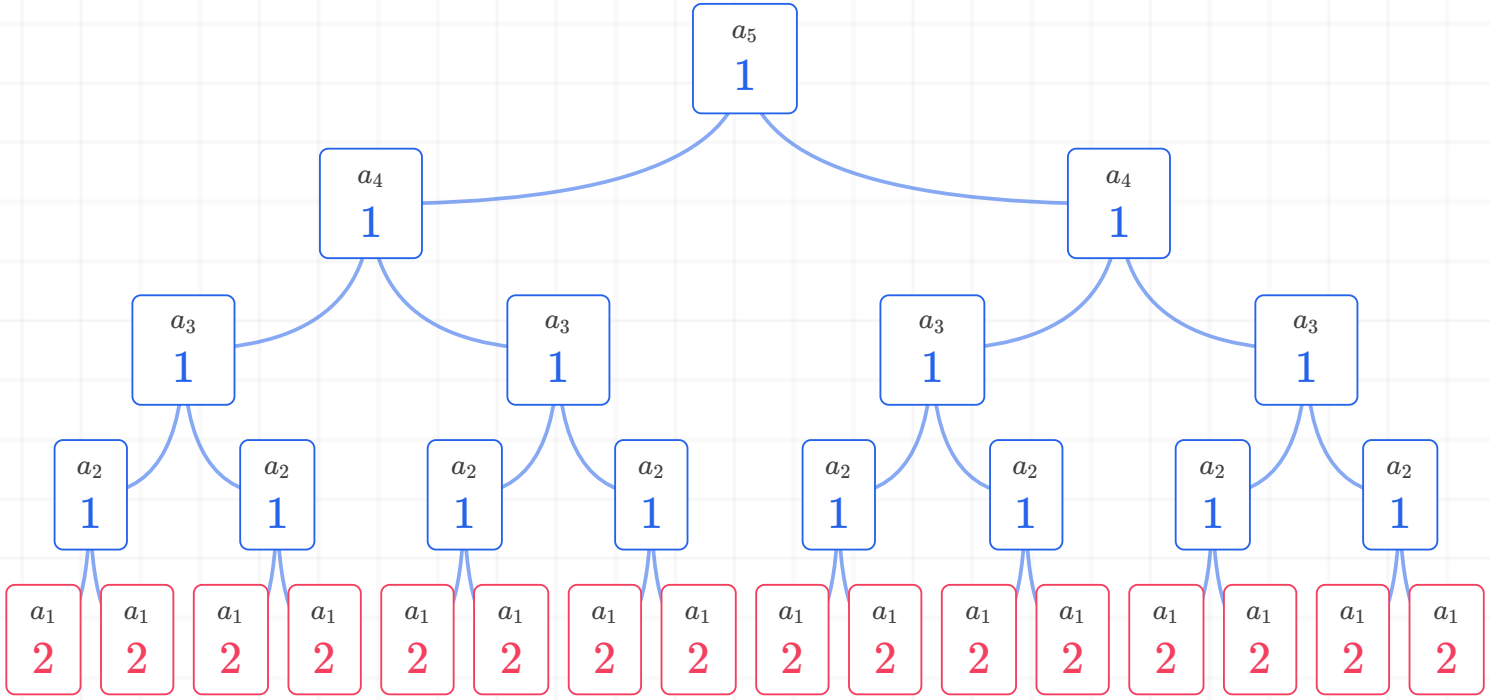
$$1 + 2 + 4 + 8 = 15$$

Sixteen leaves, each worth 2

$$16 \times 2 = 32$$

All contributions

$$a_5 = 15 + 32 = 47$$



$$a_5 = 1 + 2 + 4 + 8 + 32 = 47$$

EXERCISE 4 · CHECKPOINT · 1 MINUTE

Fill the layer counts

The root is layer 0; the base is layer h. Let $n \geq 2$.

$$a_n = 2a_{n-1} + 1, \qquad a_1 = 2$$

01 At which layer does the index reach 1?

$h = n - 1$.

02 At layer $i < h$: index, box count, and value per box?

$n - i$, 2^i, and 1.

03 At the base: box count and value per box?

2^{n-1} and 2.

CHECK TOGETHER

Count the layers, then add their contributions

Branching factor 2 gives 2^i nodes. Subproblem index $n - i$ reaches 1 after $h = n - 1$ steps.

Layer	Subproblem index	Nodes × value
$0 \le i < h$	$n - i$	$2^i \cdot 1$
h • **base**	1	$2^{n-1} \cdot 2$

$$a_n = (2^{n-1} - 1) + 2 \cdot 2^{n-1} = 3 \cdot 2^{n-1} - 1$$

EXERCISE 5 · CHECKPOINT · 45 SECONDS

Count with branching factor 1

Start at index 4.

$$a_n = a_{n-1} + 1 \quad (n \geq 2), \qquad a_1 = 2$$

01 **Links per expanding box? Steps to index 1?**

1 **link;** 3 **steps.**

02 **What is a_4?**

$a_4 = 1 + 1 + 1 + 2 = 5.$

CHECK TOGETHER

Change only the branching factor

$a_n = a_{n-1} + 1, a_1 = 2$. Branching factor 1 makes a single path.

Same subproblem size and nonrecursive part

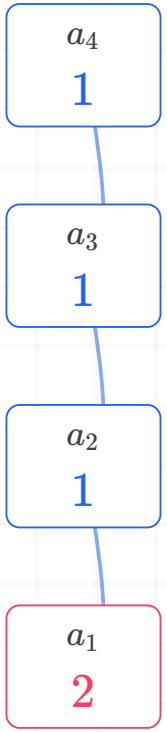
Subtract 1 from the index; write 1 until the base.

One copy at every step

$$a_4 = 1 + 1 + 1 + 2 = 5$$

For any starting index

$$a_n = (n - 1) + 2 = n + 1$$



Branching factor changes the number of nodes, even when the depth stays the same.

EXERCISE 6 · CHECKPOINT · 1 MINUTE

Follow the new subproblem-size rule

Start at index 27. Use powers of two.

$$a_n = 3a_{n/3} + 1 \quad (n \geq 2), \qquad a_1 = 2$$

01 **List the indices on one path to the base.**

$16 \to 8 \to 4 \to 2 \to 1.$

02 **How many steps? How many base boxes?**

4 steps; 16 base boxes.

03 **What is a_{16}?**

$a_{16} = 15 + 32 = 47.$

CHECK TOGETHER

Change only the subproblem size

$a_n = 2a_{n/2} + 1, a_1 = 2, n = 2^k$.
Branching factor 2; nonrecursive part 1.

Halving reaches the base

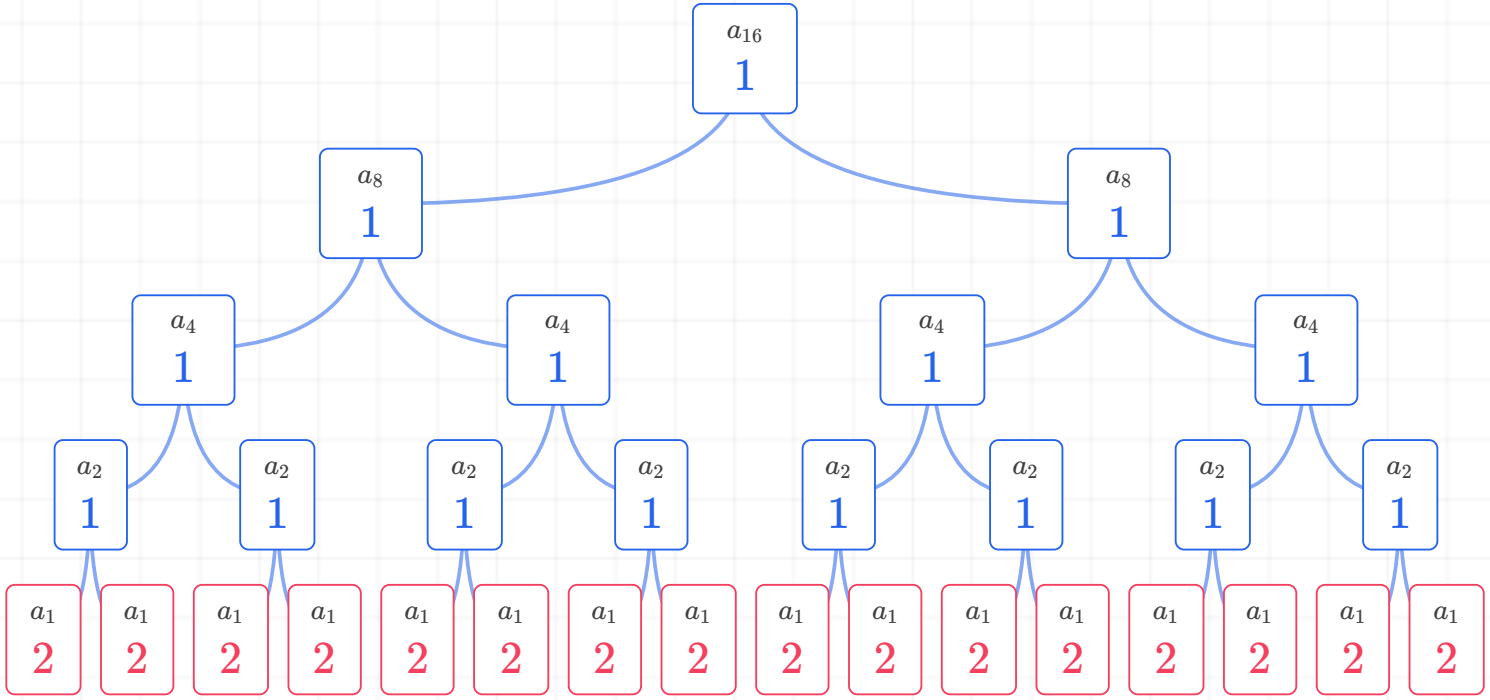
$$h = \log_2 n$$

Count the leaves

$$2^h = n$$

Add internal values and base values

$$a_n = (n-1) + 2n = 3n - 1$$



$a_{16} = 47$**: the same picture as before, reached by different subproblem sizes.**

Compare the subproblem-size rules

Fix branching factor 1, nonrecursive part 1, and base value $a_1 = 2$. Use powers of two for halving.

Subproblem size $n - 1$

$$a_n = a_{n-1} + 1 \quad \Rightarrow \quad a_n = 2 + (n - 1)$$

Subproblem size $n/2$

$$a_n = a_{n/2} + 1 \quad \Rightarrow \quad a_n = 2 + \log_2 n$$

There is one leaf in both cases. The subproblem-size rule determines the number of additions.

Evaluate the nonrecursive part at each index

$a_n = a_{n-1} + n$, $a_1 = 2$. Branching factor 1; subproblem size $n - 1$; nonrecursive part n.

At $n = 4$

$$a_4 = 4 + 3 + 2 + 2 = 11$$

Add the path for general n

$$a_n = 2 + \sum_{j=2}^{n} j = 1 + \frac{n(n+1)}{2}$$

The path is unchanged. A growing nonrecursive part changes a linear total into a quadratic one.

EXERCISE 7 · CHECKPOINT · 1 MINUTE

Use the index of each box

Start at index 4.

$$a_n = a_{n-1} + n^2 \quad (n \geq 2), \qquad a_1 = 2$$

01 **Values at indices 4, 3, 2, 1? Their sum?**

$(16, 9, 4, 2)$; $a_4 = 31$.

02 **Compared with $+n$, do connections, values, or both change?**

Only node values change.

CHECK TOGETHER

Check the nonrecursive part n^2

$a_n = a_{n-1} + n^2$, $a_1 = 2$. The branching factor and subproblem-size rule stay the same.

Check the attempted sum

$$a_4 = 16 + 9 + 4 + 2 = 31$$

Generalize the same path

$$a_n = 2 + \sum_{j=2}^{n} j^2 = 1 + \frac{n(n+1)(2n+1)}{6}$$

A squared nonrecursive part gives a cubic total along this single decrementing path.

Branching and subproblem size can balance

$a_n = 2a_{n/2} + n$, $a_1 = 2$, $n = 2^k$. Branching factor 2; subproblem size $n/2$; nonrecursive part n.

Layer i	Nodes × value	Contribution
$0 \le i < k$	$2^i \cdot (n/2^i)$	n
k • **base**	$n \cdot 2$	$2n$

$a_n = n \log_2 n + 2n$. **At** $n = 8$**:** $8 + 8 + 8 + 16 = 40$**.**

EXERCISE 8 · CHECKPOINT · 1 MINUTE

Check one row before summing the tree

Start at $n = 8$. Give each row as (box count, value per box, total).

$$a_n = 2a_{n/2} + n^2 \quad (n \geq 2), \qquad a_1 = 2$$

01 **What is the row at index 4?**

$(2, 16, 32)$.

02 **What is the base row at index 1?**

$(8, 2, 16)$.

03 **Do internal-row totals halve, stay equal, or double?**

Halve.

CHECK TOGETHER

Check the effect of a squared nonrecursive part

$a_n = 2a_{n/2} + n^2, a_1 = 2, n = 2^k$. Keep branching factor 2 and subproblem size $n/2$.

Internal layer $0 \le i < k$

$$2^i(n/2^i)^2 = n^2/2^i$$

Check the attempt at $n = 8$

$$64 + 32 + 16 + 16 = 128$$

Size 4: $2 \cdot 16 = 32$. Base row: $8 \cdot 2 = 16$.

Add the internal layers and leaves

$$a_n = n^2 \sum_{i=0}^{k-1} 2^{-i} + 2n = 2n^2$$

Internal layer totals now halve: doubling nodes does not offset quartering each contribution.